Least squares

argmax/min invariant to 1/2 but it is useful in the computations: $\partial E / \partial W = 0$ where E is quadratic

$$\begin{pmatrix} \tilde{X}\tilde{W} \\ \vdots \end{pmatrix} \text{prediction} - \begin{pmatrix} T \\ \vdots \end{pmatrix} \text{ground truth} = \text{error}$$

Column view

$$A^T \cdot A = \begin{bmatrix} c_1^T \\ \vdots \\ c_N^T \end{bmatrix} [c_1 \cdots c_N]$$

in the diagonal you find the squared norm of each column

Minimize the sum-of-squares error function $E(\tilde{W}) = \frac{1}{2} \mathrm{Tr} \left\{ (\tilde{X}\tilde{W} - T)^T (\tilde{X}\tilde{W} - T) \right\}$

closed-form solution $\tilde{W}^* = \boxed{(\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T} T$ pinv(x)·T in matlab/numpy

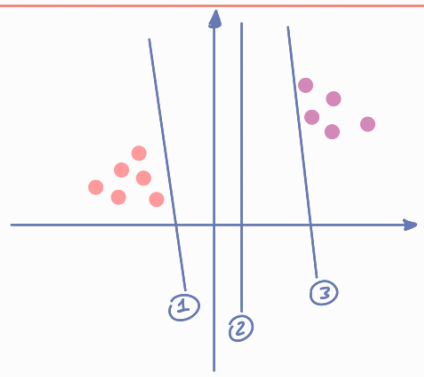
pseudo inverse of X

Once we have $\tilde{W}^*$, to classify a new instance:

$$y(x) = \tilde{W}^T \tilde{x} = \begin{pmatrix} y_1(x) \\ \vdots \\ y_N(x) \end{pmatrix} \longrightarrow k = \underset{i \in [1, \cdots, K]}{\operatorname{argmax}} \begin{pmatrix} y_1(x) \\ \vdots \\ y_N(x) \end{pmatrix}$$

Problem: not robust to outliers a.k.a. points in the dataset that possibly comes from a different distribution

Outlier: an outlier is a data point that differs significantly from other observations



1. possible solution with small ρ
2. not a possible solution with small ρ
3. possible solution with small ρ

Solution 2 should be instead preferable since it better classifies instances not in the dataset.